SMART INDIA HACKATHON 2026 · DRAFT TECHNICAL SUMMARY

Navigating a vehicle when the satellites go away

A smartphone-only dead-reckoning pipeline for GNSS outages, built on IO-VNBD and measured end to end against baselines that read no inertial data at all.

187 sessions · 149 k windows   78,982-parameter TCN   60 s outage · test split   5 September 2026

01 What the system does

When GNSS drops out — a tunnel, an urban canyon, a multi-storey car park — the vehicle has to keep knowing where it is from the phone in the cradle and nothing else. Our pipeline predicts **how far the vehicle travels each second** from a 3-second window of accelerometer and gyroscope data, then walks that distance along the road network to produce a position.

The separation matters. Distance comes from the learned model; direction comes from the map. Nothing integrates heading over the length of the outage, which is where classical inertial navigation loses its position.

SENSORS	MODEL	ROAD GRAPH	TRACKER	HARNESS
6 channels at 10 Hz, rotated into a gravity-levelled frame so only mount yaw is unknown.	Dilated TCN over a 3 s window → displacement, its uncertainty, and a stationary flag.	6.7 M-point OpenStreetMap graph, in-process. No server, no network.	Advance the predicted distance along the matched road; gyro only picks the branch at junctions.	Simulated outages at 10 / 30 / 60 s, scored against GPS truth.

02 Data and labels

IO-VNBD, 187 unique sessions across four vehicles and three phones. Splits are assigned by vehicle and phone — never randomly — because windows overlap by nine of their ten seconds and a random split would put near-duplicates on both sides.

The label is displacement over the following second, taken from cumulative geodesic distance between GPS fixes. It is an integral of position, never a differentiated speed: differentiating a 1 Hz quantised signal to make a 10 Hz reference amplifies its noise, and we measured the ceiling that imposes — even a *perfect* ECU accelerometer only reaches r = 0.12–0.58 against such a reference, against 0.93 for clean wheel speed.

Two standing guards protect the labels. Any window implying more than 60 m/s is rejected, and any session whose fix-to-fix path length disagrees with the integral of its own speedometer by more than a set margin is refused outright — that caught four sessions whose GPS positions were inflated by 2.1× to 3.5×.

Windows per split, after both guards. Displacement over 1 s is numerically the mean speed, so the label is its own regime key.

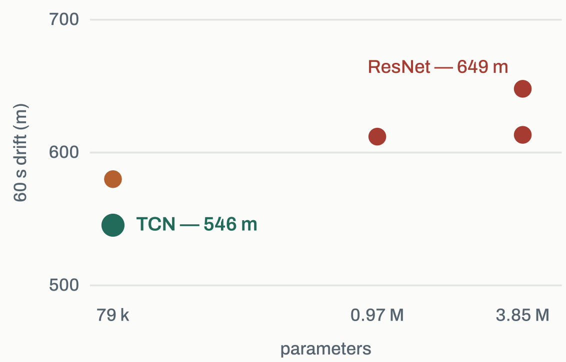
SPLIT	SESSIONS	WINDOWS	MEAN LABEL	REJECTED
train	47	83,238	13.85 m	0.11%
validate	7	38,117	24.63 m	0.05%
test	4	27,937	19.67 m	0.00%

03 The model

A convolutional stem feeding four causal dilated residual blocks (dilations 1, 2, 4, 8), global-average pooled into three heads: displacement with a predicted log-variance, a stationary logit, and a yaw-rate output. **78,982 parameters** — 49× smaller than the 1-D ResNet it replaced, and better on every measure.

Training is two-phase: a plain-MSE warm-up so the variance head cannot buy loss reduction before the mean is calibrated, then the full multi-task objective. Early stopping is on **60-second drift through the outage harness**, not windowed error — small, consistently-signed per-window errors accumulate into large drift, so windowed MAE can improve while the deployable quantity gets worse.

SMALLER MODEL, LOWER DRIFT



Every ResNet variant we trained sits above every TCN variant, at up to 49× the parameter count.

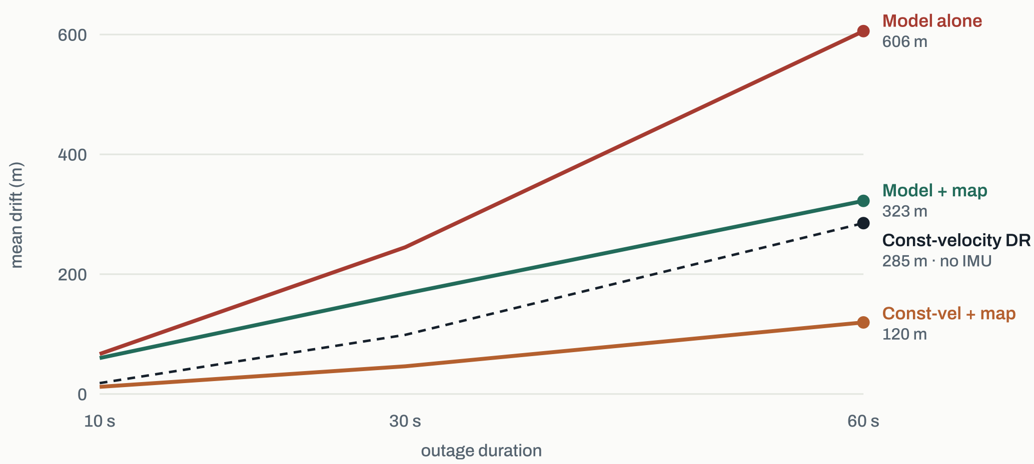
04 Validating the measuring instrument first

Before trusting any learned result we reproduced a published one. Onyekpe et al. (2021) report classical INS dead reckoning on IO-VNBD across five driving scenarios; our implementation reproduces the sequence counts almost exactly (9/9, 12/11, 15/13, 15/17, 35/40) and lands within 30% on three of five, with the difficulty ordering preserved — motorway lowest, roundabout highest.

That ordering test earned its keep immediately: it caught a defect in our own metric. Reading CRSE as the scalar sum of per-second distance errors *inverts* the published ordering, making the motorway the hardest scenario. It has to be the norm of the accumulated error *vector* — a scalar sum cannot see heading error at all. Magnitudes alone would never have exposed that.

05 Results

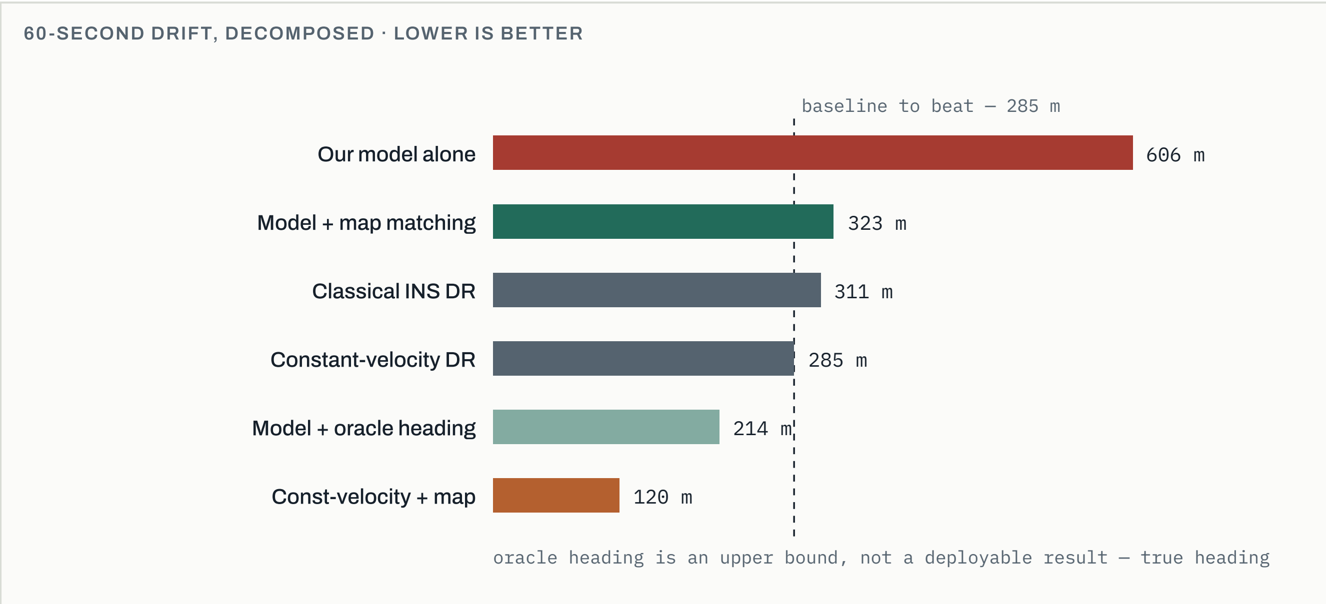
MEAN POSITION DRIFT BY OUTAGE LENGTH · TEST SPLIT, 417 OUTAGES



Map matching roughly halves drift for every predictor, and the benefit grows with outage length — the signature of removing accumulated heading error. The dashed line is the baseline that reads no inertial data at all.

Where the remaining error actually lives

Substituting oracle GPS heading into the model — changing nothing else — takes 60-second drift from 606 m to 214 m. That single substitution accounts for roughly two-thirds of the gap, and it puts the model *ahead* of the no-IMU baseline. The displacement estimate is sound; heading is what costs us.



06 Honest position

> **The learned model does not yet beat a baseline that reads no IMU at all.** Map-matched, we are at 323 m against 120 m for map-matched constant-velocity dead reckoning. That is the number to state plainly.

What the diagnostics establish is *why*, which is a more useful position than a single figure. Three findings, each measured rather than assumed:

- **The dominant cue is vehicle- and surface-dependent.** Vertical acceleration variance rises monotonically with speed on train and test, but *inverts* on validate — the Renault/Moto G7/French-motorway pairing simply does not produce proportional vibration at speed. Removing that channel doubles error everywhere, so it is not an incidental feature the model could route around.
- **The learned yaw head is worse than the raw gyro** on held-out test — 647 m against 939 m. It is disabled in the shipped configuration. Three of four heads earn their place; one does not.
- **Mount orientation is not identifiable during an outage.** Three independent attempts failed at the same point, and measurement explains it: the mount yaw estimate moves 13–36° *within* a session, 5.5–11× the estimator's own noise.

Several plausible fixes were tried and did not work: bucket reweighting, post-hoc calibration, despiking, ZUPT gating, a non-holonomic EKF. Each is recorded with the measurement that ruled it out, so nobody repeats them.

07 What we would do next

- **Heading, not displacement.** The oracle bound says solving heading is worth roughly 390 m of the gap. Road-network priors over a longer horizon are the obvious route, and the in-process graph already supports it.
- **Training data spanning more road surfaces**, to break the dependence on a vibration cue that does not transfer between vehicles.
- **Drop the pyproj dependency** — it wraps a C library and is the remaining blocker to embedding the matcher on a phone, now that the routing server is gone.

Map matching runs fully in-process against a cached road graph: no routing server, no HTTP, no network at inference. Scored side by side against the OSRM-backed path, the two agree to within 2.3 m of mean 60-second drift, with the in-process backend marginally ahead.

Draft · figures from results/mapmatch.csv, results/run_comparison.csv and results/stage0_validation.csv · 165 tests passing · full negative-result log in results/findings.md